$$1\text{-}p = p$$

$$1 = 2p$$

$$p = \frac{1}{2}$$

$$q = \frac{1}{2}$$

Hence the given Binomial distribution is symmetrical.

Example 10.7

From a pack of 52 cards 4 cards are drawn one after another with replacement. Find the mean and variance of the distribution of the number of kings.

Solution:

Success X=event of getting king in a draw

p=probability of getting king in a single trial

$$p = \frac{4}{52}$$

$$= \frac{1}{13}$$

This is constant for each trial.

Hence, it is a binomial distribution with $n=4$, $p = \frac{1}{13}$ and $q = \frac{12}{13}$

$$\text{Mean} = np = 4\times\frac{1}{13} = \frac{4}{13}$$

$$\text{Variance} = npq = \frac{4}{13}\times\frac{12}{13} = \frac{48}{169}$$

Example 10.8

In a street of 200 families, 40 families purchase the Hindu newspaper. Among the families a sample of 10 families is selected, find the probability that

(i) Only one family purchase the news paper

(ii) No family purchasing

(iii) Not more than one family purchase it

Solution:

$X \sim B(n, p)$

$P(X = x) = nC_x\, p^x\, q^{n-x}$, x= 0,1,2,...n